



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 1 – Fundamentals of OOP
Topic –1.9 Class, Object, Class Scope, Accessing class members and Controlling access to members

Prof. V.N.Nirgude
Assistant Professor

E-mail :

nirgudevikascomp@sanjivani.org.in

Contact No: 9975240215



Class

- Class: A Class is a user defined data type to implement an object.
- A Class is a combination of data and functions.
- Data is called as data members and functions are called as member functions.



Class definition

- A class definition begins with the keyword *class*.
- The body of the class is contained within a set of braces, { } ;
(notice the semi-colon).

```
class class_name
{
    ....    Class body
           (data member +methods)
    ....
};
```



Data member or member functions may be public, private or protected.

▼ **Access specifiers:**

- public:
 - These can be accessed outside the class directly.
 - The public stuff is *an interface*.
- private:
 - Accessible only to member functions of a class
 - Private members and methods are for internal use only.
 - Private means data members and member functions can't be used outside the class.

protected:

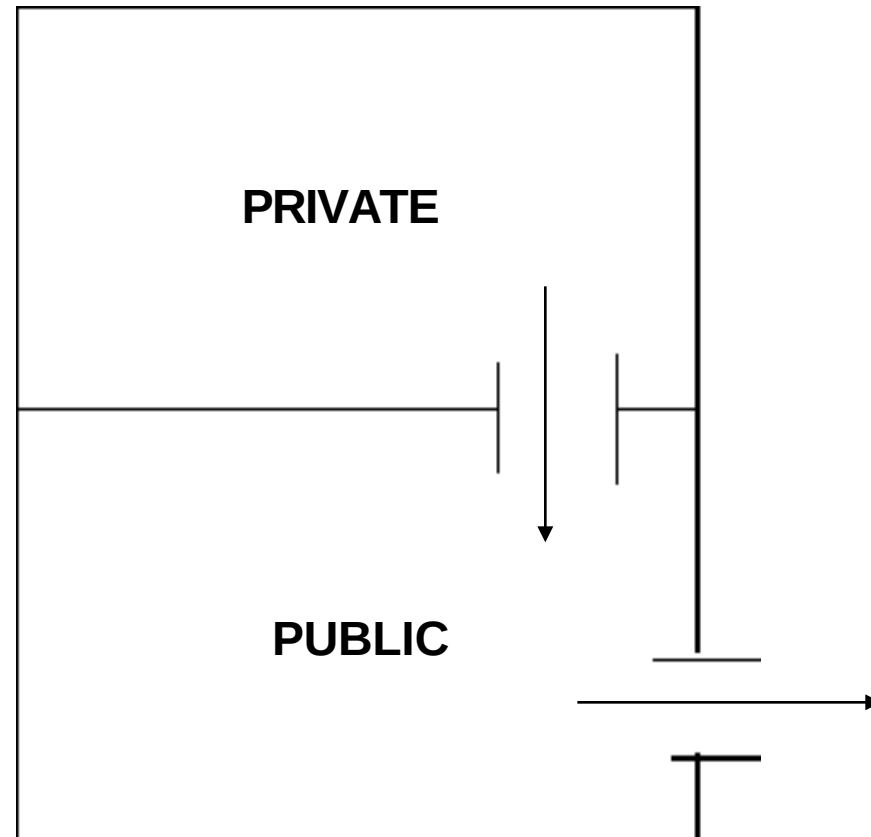
- Protected means data member and member functions can be used in the same class and its derived class (at one level of inheritance)



Within the body, the keywords *private:* and *public:* specify the access level of the members of the class.

The default access specifier is private:

- Usually, the data members of a class are declared in the *private:* section and the member functions are in *public:* section of the class.



```
class class_name
```

```
{
```

```
    private:
```

```
        ...
```

```
        ...
```

```
        ...
```

```
    public:
```

```
        ...
```

```
        ...
```

```
};
```

private members or methods

Public members or methods



Accessing Class Members

- v Operators to access class members
 - v Dot member selection operator (.)
 - v Object
 - v Reference to object
 - v Arrow member selection operator (->)
 - Pointers



Object

- ✓ An object is an instance of a class.
- ✓ An object is a class variable.
- ✓ Every object have a state which is represented by the values of its attributes. These state are changed by function which applied on the object.
- ✓ Once an object of a certain class is instantiated, a new memory location is created for it to store its data members and code.



Memory Allocation of an Object

class student

{

int rollno;

char name[20];

int marks;

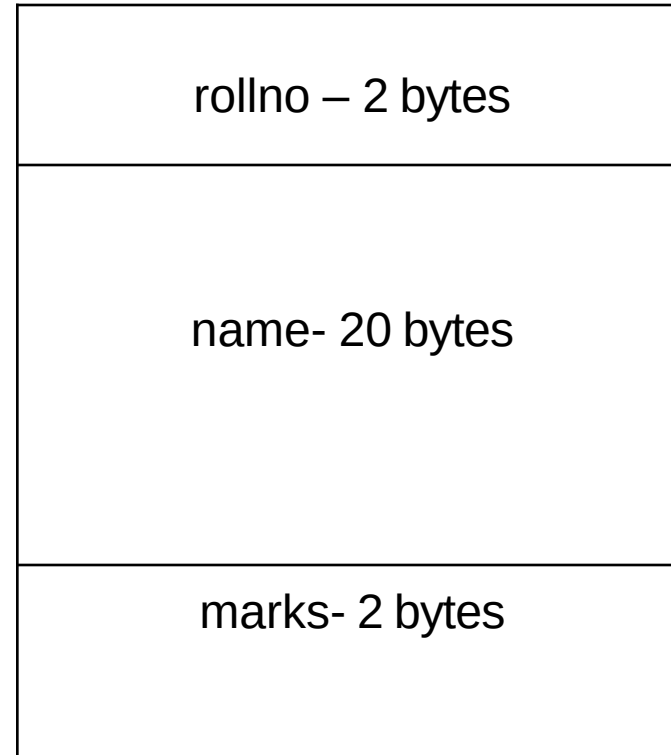
};

void main()

{

student s;

}



Total memory = 24 bytes
object-s



Member Functions Defining Inside the Class

```
#include<iostream>
class student
{
    int rollno;
    char name[20];
public:
    void getdata()
    {
        cout<<"Enter the roll no.:";

        cin>>rollno;

        cout<<"Enter the name:";
        gets(name);
    }
```

```
        void putdata()
        {
            cout<<"Your roll no:"<<rollno;
            cout<<"Your name:";
            puts(name);
        }
};
int main()
{
    student s;
    s.getdata();
    s.putdata();
    return 0;
}
```



Member Functions Defining Outside the Class

```
#include<iostream>
```

```
class student
```

```
{
```

```
    int rollno;
```

```
    char name[20];
```

```
public:
```

```
    void getdata();
```

```
    void putdata();
```

```
};
```

```
void student :: getdata()
```

```
{
```

```
    cout<<"enter the rollno.:";
```

```
    cin>>rollno;
```

```
    cout<<"enter the name:";
```

```
    gets(name);
```

```
}
```

```
void student :: putdata()
```

```
{
```

```
    cout<<"Your rollno:"<<rollno;
```

```
    cout<<"Your name:";
```

```
    puts(name);
```

```
}
```

```
int main()
```

```
{
```

```
    student s;
```

```
    s.getdata();
```

```
    s.putdata();
```

```
    return 0;
```

```
}
```



Thank You..